YOU'RE AN ALIEN EXPLORER entering the Solar System after a long journey through the blackness of interstellar space. You examine the planets of this humdrum star from afar—a pretty handful, some gray, some blue, some red, some yellow. You're interested in what kinds of worlds these are, whether their environments are static or changing, and especially whether there are life and intelligence. You have no prior knowledge of the Earth. You've just discovered its existence.

There's a galactic ethic, let's imagine: Look but don't touch. You can fly by these worlds; you can orbit them; but you are strictly forbidden to land. Under such constraints, could you figure out what the Earth's environment is like and whether anyone lives there?

As you approach, your first impression of the whole Earth is white clouds, white polar caps, brown continents, and some bluish substance that covers two thirds of the surface. When you take the temperature of this world from the infrared radiation it emits, you find that most latitudes are above the freezing point of water, while the polar caps are below freezing. Water is a very abundant material in the Universe; polar caps made of solid water would be a reasonable guess, as well as clouds of solid and liquid water.

You might also be tempted by the idea that the blue stuff is enormous quantities—kilometers deep—of liquid water. The suggestion is bizarre, though, at least as far as *this* solar system is concerned, because surface oceans of liquid water exist nowhere else. When you look in the visible and near-infrared spectrum for telltale signatures of chemical composition, sure enough you discover water ice in the polar caps, and enough water vapor in the air to account for the clouds; this is also just the right amount that must exist because of evaporation if the oceans are in fact made of liquid water. The bizarre hypothesis is confirmed.

The spectrometers further reveal that the air on this world is one fifth oxygen, O_2. No other planet in the Solar System has anything close to so much oxygen. Where does it all come from? The intense ultraviolet light from the Sun breaks water, H_2O, down into oxygen and hydrogen, and hydrogen, the lightest gas, quickly escapes to